Kwame Nkrumah University of
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ME 255 – Strength of Materials

Lecture 3: Torsion in Shafts

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Lecture objective

At the end of this lecture, you will understand the principles of

- Shear strain due to torsional load for circular sections
- Shear stress due to torsional load for circular sections
- Power generation by transmission shafts
- Shear stress due to torsional load for hollow, thin-walled shafts
- Torsional tests

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Introduction

- A shaft is said to be in torsion, when equal and opposite torques are applied at the two ends of the shaft.
- The torque is equal to the product of the force applied (tangentially to the ends of a shaft) and radius of the shaft.
- Due to the application of the torques at the two ends, the shaft is subjected to a twisting moment. This causes the shear stresses and shear strains in the material of the shaft.
- Torsion refers to the twisting of long members.

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Introduction

- Torsion can occur with members of any cross-sectional shape, but the most common is the circular shaft.
- Torsion of circular shafts are considered because a circular cross section is an efficient shape for resisting torsional loads.
- Circular shafts are commonly used to transmit power in rotating machinery.
- Another fairly common shaft configuration, which has a simple solution, is the hollow, thin-walled shaft.

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Assumptions on circular shafts under torsion

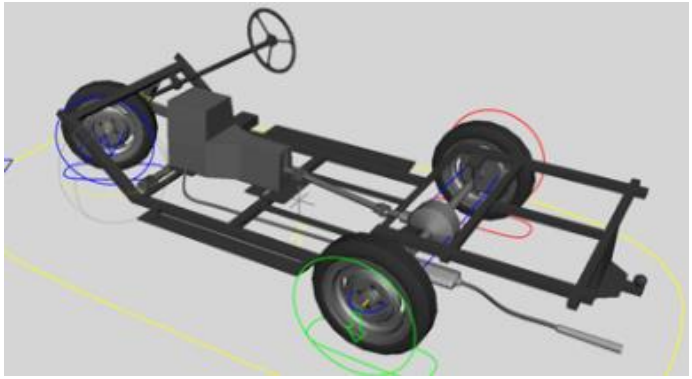
- The material of the shaft is uniform throughout.
- The twist along the shaft is uniform.
- The shaft is of uniform circular section throughout.
- During deformation, the cross sections are not distorted in any manner.
- All radii which are straight before twist remain straight after twist.
- In addition, the **length L** of the shaft remains constant.

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Torsion in Everyday Life



Fitting a ceiling light



Automobiles



Spanner



Opening a Jar



Screws



Drying wet clothes

Circular Shaft - Strain

- **Shear strain** is given by

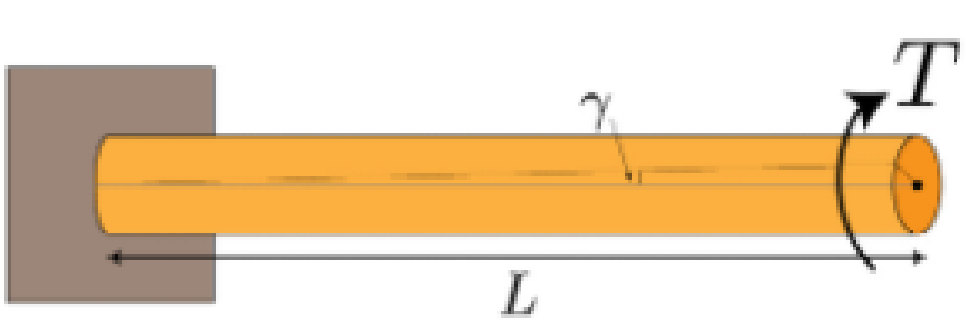
$$\gamma = \frac{\rho\theta}{L}$$

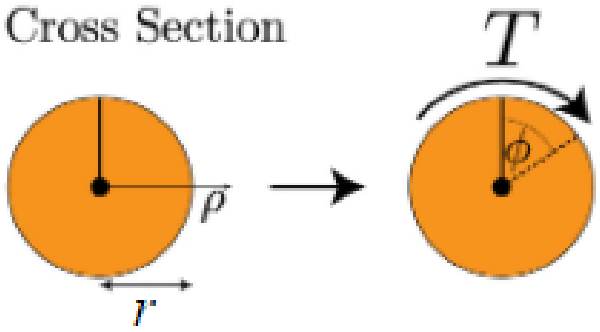
- Where ρ is distance along the cross section (Radius of shaft)

  θ is angle of twist

  L is length of the rod

- $\rho = 0$ at the centre and $\rho = r$ at the outer edge of the rod.





Circular Shaft

- From the shear strain formula, we can learn that:
  i. the more angle of twist, the larger the shear strain
  ii. the shear strain is not uniform along the cross section. This is the big difference between axial-loaded structures and torque-loaded ones. It is zero at the centre of the twisted rod, and is at a maximum value at the edge of the rod.
  iii. the longer the rod, the smaller the shear strain

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Circular shaft – Shear Stress

- Shear stress is given by

$$\tau = \frac{Tr}{J}$$

- Where r is distance along the cross section

  T is torque

  J is second polar moment of area

- For solid shafts $J = \frac{\pi}{2} r^4$

- For hollow shafts $J = \frac{\pi}{2}(r_o{}^4 - r_i{}^4)$

  where r_o is outer radius and r_i is inner radius

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Circular shaft – Hooke's law

- Applying Hooke's law, the modulus of rigidity (G) is given by

$$G = \frac{shear\ stress\ induced}{shear\ strain\ produced}$$

- Thus

$$\mathbf{G = \frac{Tr}{J} \div \frac{r\theta}{L}}$$

- The angle of twist can be calculated as

$$\boldsymbol{\theta = \frac{TL}{GJ}}$$

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Maximum Torque Transmitted by a Circular Solid Shaft

- The maximum torque transmitted by a circular solid shaft, is obtained from the maximum shear stress induced at the outer surface of the solid shaft.
- Consider a shaft subjected to a torque T, the maximum torque transmitted by the shaft is given by

$$T = \frac{\pi}{16}\tau D^3$$

- Where T is the torque, τ is maximum shear stress and D is diameter of the shaft.

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Example 1

- A solid shaft of 150 mm diameter is used to transmit torque. Find the maximum torque transmitted by the shaft if the maximum shear stress induced to the shaft is 45 N/mm^2.

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Solution

Given : Diameter of the shaft, D = 150 mm

Maximum shear stress, $\tau = 45\ \mathrm{N/mm^2}$

Let T = Maximum torque transmitted by the shaft.

$$T = \frac{\pi}{16}\tau D^3$$

$$T = \frac{\pi}{16} \times 45 \times 150^3$$

$$\boldsymbol{T = 29820586\ Nmm}$$

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Example 2

- The shearing stress of a solid shaft is not to exceed 40 N/mm^2 when the torque transmitted is 20000 N-m. Determine the minimum diameter of the shaft.

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Solution

Maximum shear stress, $\tau = 40\ \mathrm{N/mm^2}$

Torque transmitted, $T = 20000$ N-m $= 20000 \times 10^3\ \mathrm{N-mm}$

Let D = Minimum diameter of the shaft in mm.

$$T = \frac{\pi}{16}\tau D^3$$

$$D = \left(\frac{16T}{\pi\tau}\right)^{1/3}$$

$$D = \left(\frac{16 \times 20000 \times 10^3}{\pi \times 40}\right)^{1/3}$$

$$D = \mathbf{136.2\ mm}$$

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Power transmission

- One of the most common examples of torsion in engineering design is the power generated by transmission shafts.
- The power per rotation of a circular rod is equal to the applied torque times the frequency of rotation and is given by

$$P = \frac{2\pi NT}{60} = \omega T$$

Where **N** = r.p.m. of the shaft

T = Mean torque transmitted in N-m

ω = Angular speed of shaft.

P = Power in Watts

Example 3

The hollow circular steel shaft has an inner diameter of 25 mm, an outer diameter of 50 mm, and a length of 600 mm. It is fixed at the left end and subjected to a torque of 1400 N-m. Find the maximum shear stress in the shaft and the angle of twist at the right end. Take G= 84 GPa.

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Solution

Data Given

d_i = 25 mm; d_o = 50 mm; L = 600 mm; T = 1400 N-m

$$J = \frac{\pi}{2}\left(r_o^4 - r_i^4\right) = 575E-03m^4$$

$$\tau_{\phi z} = \frac{Tr_o}{J} = 60.8MPa$$

$$\theta = \frac{TL}{GJ} = 0.0174\,rad$$

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Example 4

A shaft 0.7m long is subjected to a torque 1200Nm. Calculate the shear stress and angle of twist if

a) it is a solid shaft with diameter 50mm

b) hollow shaft with outer diameter 50mm and inner diameter 30mm

Take G=90GPa

Solution (a)

Important values to use are D=0.05 m, L=0.7 m, T=1200 Nm, G=90 GPa

$$J = \frac{\pi r^4}{2} = \frac{\pi \times 0.025^4}{2} = 613.59 \times 10^{-9}\ m^4$$

$$\tau_{max} = \frac{Tr}{J} = \frac{1200 \times 0.025}{613.59 \times 10^{-9}} = 48.89\ MPa$$

$$\theta = \frac{TL}{GJ} = \frac{1200 \times 0.7}{90 \times 10^9 \times 613.59 \times 10^{-9}} = 0.0152\ radian$$

Converting to degrees: $\theta = 0.0152 \times \frac{180}{\pi} = 0.871°$

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Solution (b)

- Important values to use are Do=0.05 m, Di=0.03m, L=0.7 m, T=1200 Nm, G=90 GPa

$$J = \frac{\pi({r_o}^4 - {r_i}^4)}{2} = \frac{\pi \times (0.025^4 - 0.015^4)}{2} = 534.07 \times 10^{-9}\ m^4$$

$$\tau_{max} = \frac{Tr}{J} = \frac{1200 \times 0.025}{534.07 \times 10^{-9}} = 56.17\ MPa$$

$$\theta = \frac{TL}{GJ} = \frac{1200 \times 0.7}{90 \times 10^9 \times 534.07 \times 10^{-9}} = 0.0175\ radian$$

Converting to degrees: $\theta = 0.0175 \times \frac{180}{\pi} = 1°$

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Example 5

A shaft is made from tube. The ratio of the inside diameter to the outside is 0.6. The material must not experience a shear stress greater than 500 kPa. The shaft must transmit 1.5 MW of mechanical power at 1500 rev/min. Calculate the shaft diameters.

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Solution

The important quantities are $P = 1.5 \times 10^6$ Watts,

$\tau = 500 \times 10^3$ Pa, $N = 1500$ rev/min and $d = 0.6D$.

$N = 1500 \text{ rev/min} = 1500/60 = 25 \text{ rev/s}$ $\qquad P = 2\pi N T$

$$\text{hence } T = \frac{P}{2\pi N} = \frac{1.5 \times 10^6}{2\pi \times 25} = 9549.3 \text{ Nm}$$

$$J = \frac{\pi\left(D^4 - d^4\right)}{32} = \frac{\pi\left\{D^4 - (0.6D)^4\right\}}{32} = \frac{\pi\left\{D^4 - 0.36D^4\right\}}{32} = 0.08545D^4$$

$$\frac{T}{J} = \frac{\tau}{R} = \frac{2\tau}{D} \quad \text{hence} \quad \frac{9549.3}{0.08545D^4} = \frac{2 \times 500 \times 10^3}{D} = \frac{9549.3}{0.08545 \times 2 \times 500 \times 10^3} = \frac{D^4}{D} = D^3$$

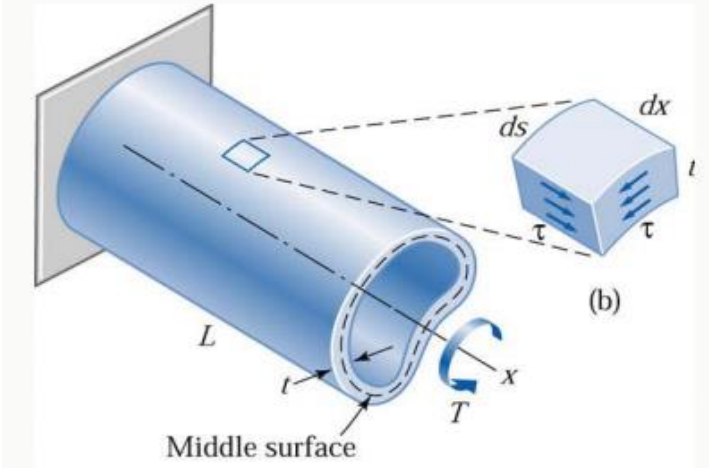
$D^3 = 0.11175$ $\qquad D = \sqrt[3]{0.11175} = 0.4816 \text{ m} = 481.6 \text{ mm}$ $\qquad d = 0.6D = 289 \text{ mm}$

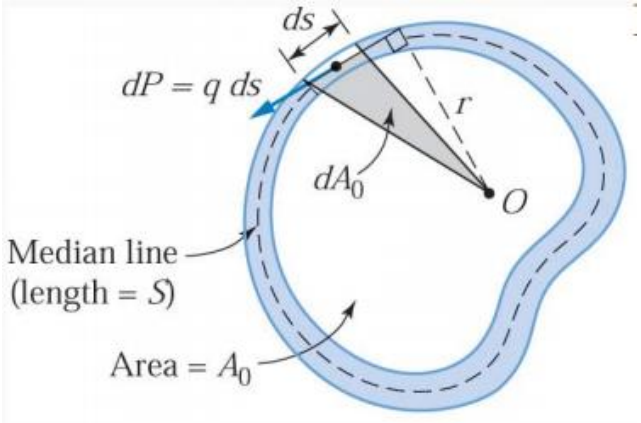
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Hollow, Thin- Walled Shafts

- Figures below show the cross-section of the thin walled tube of non-constant thickness.
- The definition of the ***shear*** can be expressed as $\boldsymbol{\tau = \frac{T}{2A_o t}}$
- ***Angle of twist*** is given by $\boldsymbol{\theta = \frac{TLS}{4GA_o{}^2 t}}$
- Where A_o is the area enclosed by the median line

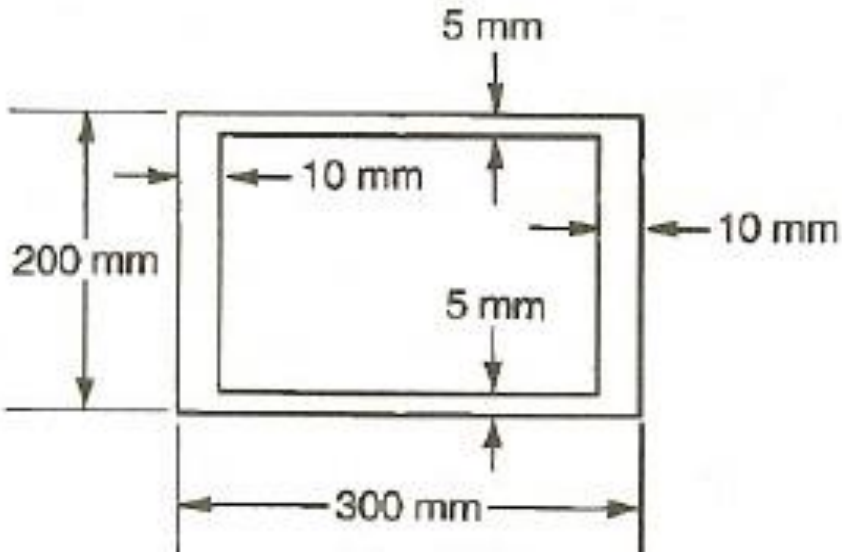
t is the wall thickness; T is the torque an S is length of the median line





Example 6

A torque of 10 kN-m is applied to the thin-walled rectangular steel shaft whose cross-section is shown in Fig. below. The shaft has wall thickness of 5 mm and 10 mm. Find the maximum shear stress in the shaft.



Solution

Data Given

T = 10 kN-m

Solve

A_o = (200-5)(300-10) = 56,550 mm^2

The maximum shear stress will occur in the thinnest section, so t = 5 mm.

$$\tau = \frac{T}{2A_o t} = \frac{10000}{(0.05655)(0.005)} = 17.68\ MPa$$

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Torsional Test

- Torsional testing is to determine Shear Modulus of Elasticity (G) of some Steel, Aluminum and Brass circular shafts and develop a relationship among the Torque (T) and Clamping length (L) and the angle of twist (θ).
- The test set-up is as shown in Fig. below



Torsional Testing

The stress

$$\tau = \frac{Tr}{J}$$

The angle θ, of twist is

$$\theta = \frac{TL}{GJ}$$

The strain,

$$\gamma = \frac{\rho\theta}{L}$$

Example 7

A bar of metal 25 mm in diameter is tested on a length of 250 mm. A torsion test gave the following results (Table below).

Torque (kNm)	**0.051**	**0.152**	**0.253**	**0.354**
Angle of twist (θ)	0.24	0.71	1.175	1.642

Represent these results in graphical form and hence determine the modulus of rigidity, G, for the metal.

Solution

- From

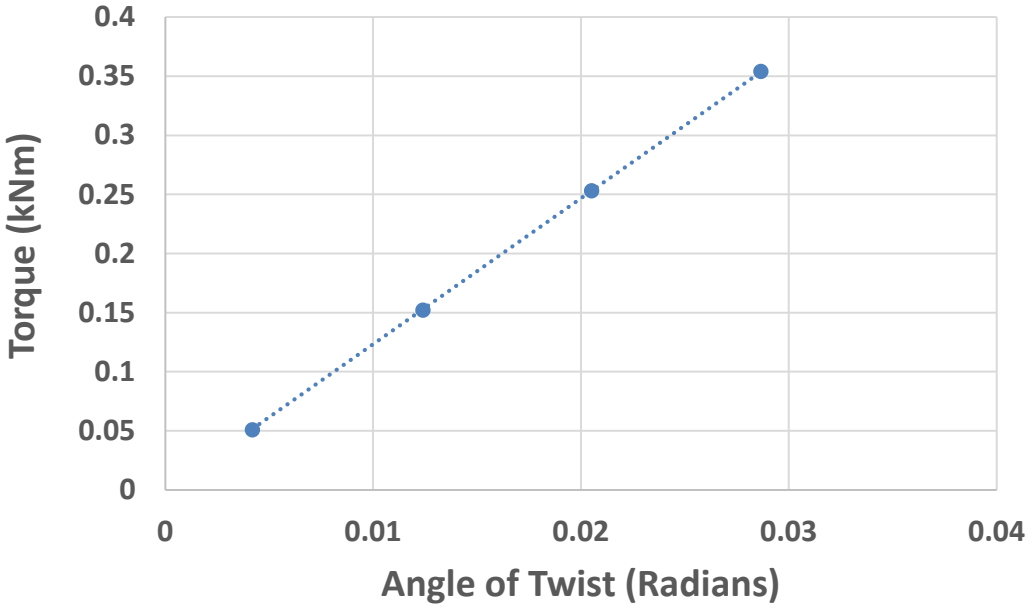
$$\theta = \frac{TL}{GJ}$$

$$T = \frac{GJ}{L}\theta$$

Torque	Angle of Twist	
kNm	Degrees	Radians
0.051	0.240	0.004189
0.152	0.710	0.012393
0.253	1.175	0.02051
0.354	1.642	0.028662

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Solution



Plotting of T against θ,

The slope is $\frac{GJ}{L}$

Hence,

$$G = \frac{slope\ x\ L}{J}$$

But, $$J = \frac{\pi}{32} d^4 = \frac{\pi}{32} (25)^4$$

$$\Rightarrow J = 38349.52\ mm^4$$

From the plot, the slope is 12.335×10^6 N/mm^2

Solution

This implies

$$G = \frac{12.335\ x\ 10^{6}(250)}{38349.52} = 80411\ N/mm^{2}$$

Hence, the modulus of rigidity, *G is 80411 N/mm*^{2}

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THANK YOU